

4.1 Noncentral chi-squared, t and F distributions

A regression analysis reports a handful of numbers whose sampling behaviour decides everything: the residual sum of squares, the reduction in it when a term is added, the ratio of the two. Each is a function of the response vector of the form

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} Y_i Y_j,$$

a **quadratic form** in $\mathbf{Y}$. The sample variance is one, with $\mathbf{A} = (n-1)^{-1}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)$. So is the squared length $\|\mathbf{P}\mathbf{Y}\|^2 = \mathbf{Y}^\top \mathbf{P} \mathbf{Y}$ of any symmetric idempotent image of $\mathbf{Y}$, and this is how the sums of squares of [Chapter 6](#) will arise.

Since $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ is a scalar, it equals its own transpose $\mathbf{Y}^\top \mathbf{A}^\top \mathbf{Y}$, and so also $\mathbf{Y}^\top \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) \mathbf{Y}$. Replacing $\mathbf{A}$ by its symmetric part changes nothing, and *throughout this chapter the matrix of a quadratic form is symmetric*. Several formulas below, the variance formula among them, are false for a non-symmetric $\mathbf{A}$ (Exercise (A1)).

When $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ and $\mathbf{A}$ is symmetric idempotent, $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$ has a chi-squared distribution, which is the null distribution of a test. Under an alternative the mean is not zero, and a slightly larger family appears. We start with that family.

4.1.1 Noncentral chi-squared

Recall that $\chi^2(r)$, the chi-squared distribution with r degrees of freedom, is the law of $Z_1^2 + \dots + Z_r^2$ for independent standard normal Z_i . It is the gamma distribution with shape $r/2$ and scale 2, with density

$$g_r(x) = \frac{x^{r/2-1} e^{-x/2}}{2^{r/2} \Gamma(r/2)}, \quad x > 0, \quad (4.1.1)$$

mean r , variance $2r$ and moment generating function $(1-2t)^{-r/2}$ for $t < 1/2$. By convention $\chi^2(0)$ is the point mass at zero. Everything about the noncentral version follows from one computation.

Lemma 4.1.1

Let $W \sim N(0, 1)$ and $\nu \in \mathbb{R}$. For $t < 1/2$,

$$\mathbb{E}[e^{t(W+\nu)^2}] = (1-2t)^{-1/2} \exp\left(\frac{\nu^2 t}{1-2t}\right).$$

Proof. With $u = w + \nu$ the expectation is $(2\pi)^{-1/2} \int \exp\{tu^2 - (u-\nu)^2/2\} du$. Completing the square in the exponent,

$$tu^2 - \frac{1}{2}(u-\nu)^2 = -\frac{1-2t}{2} \left(u - \frac{\nu}{1-2t}\right)^2 + \frac{\nu^2}{2} \left(\frac{1}{1-2t} - 1\right).$$

The last term equals $\nu^2 t / (1 - 2t)$. The remaining Gaussian integral is $(2\pi)^{1/2} (1 - 2t)^{-1/2}$ when $1 - 2t > 0$.

■

Definition 4.1.1 — Noncentral chi-squared

Let $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I}_r)$. The distribution of $\mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$ is the **noncentral chi-squared distribution** with r degrees of freedom and **noncentrality** $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu} = \|\boldsymbol{\mu}\|^2$, written $\chi^2(r, \gamma)$.

The definition hides a claim: the law of $\|\mathbf{Z}\|^2$ depends on the mean vector $\boldsymbol{\mu}$ only through its length. [Theorem 4.1.2](#) proves it. Intuitively, a rotation $\mathbf{Q}\mathbf{Z}$ of a spherical normal is again spherical normal, with mean $\mathbf{Q}\boldsymbol{\mu}$, and it has the same length as $\mathbf{Z}$, so every mean vector of a given length gives the same answer.

Remark (Convention). In this book the noncentrality is $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu}$, with no factor $\frac{1}{2}$. Then $E \chi^2(r, \gamma) = r + \gamma$, which is easy to remember, and γ agrees with the parameter `nc` of `scipy.stats.ncx2`. Several texts, including Rencher and Schaalje (2008) and Christensen (2020), use $\lambda = \boldsymbol{\mu}^\top \boldsymbol{\mu} / 2$ instead. When reading them, replace their λ by $\gamma/2$. The same convention applies to the noncentral F distribution below.

Theorem 4.1.2 — Properties of the noncentral chi-squared

Let $U \sim \chi^2(r, \gamma)$.

- (a) The moment generating function is

$$M_U(t) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right), \quad t < \frac{1}{2}, \quad (4.1.2)$$

so the law depends on $\boldsymbol{\mu}$ only through γ , and $\chi^2(r, 0) = \chi^2(r)$.

- (b) $E(U) = r + \gamma$ and $\text{Var}(U) = 2r + 4\gamma$.
(c) If $U_1, \dots, U_k$ are independent with $U_i \sim \chi^2(r_i, \gamma_i)$, then $\sum_i U_i \sim \chi^2(\sum_i r_i, \sum_i \gamma_i)$.
(d) (*Poisson mixture.*) Let $K \sim \text{Poisson}(\gamma/2)$ and, given $K = k$, let $V \sim \chi^2(r + 2k)$. Then $V \sim \chi^2(r, \gamma)$. Consequently U has density

$$f_{r,\gamma}(x) = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} g_{r+2k}(x), \quad x > 0. \quad (4.1.3)$$

Proof.

- (a) With $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$, the Z_i are independent, and $Z_i = W_i + \mu_i$ with $W_i \sim N(0, 1)$. By [Lemma 4.1.1](#),

$$E e^{t\mathbf{Z}^\top \mathbf{Z}} = \prod_{i=1}^r (1 - 2t)^{-1/2} \exp\left(\frac{\mu_i^2 t}{1 - 2t}\right) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right).$$

A moment generating function that is finite on an open interval around zero de-

termines the distribution, so the law depends only on (r, γ) . Setting $\gamma = 0$ gives the central mgf.

- (b) The cumulant generating function is $K(t) = \log M_U(t) = -\frac{r}{2} \log(1 - 2t) + \gamma t / (1 - 2t)$. Differentiating, $K'(t) = r(1 - 2t)^{-1} + \gamma(1 - 2t)^{-2}$ and $K''(t) = 2r(1 - 2t)^{-2} + 4\gamma(1 - 2t)^{-3}$. The mean and variance are $K'(0)$ and $K''(0)$.
- (c) By independence the mgf of the sum is the product of the mgfs [Eq. 4.1.2](#), which has the same form with $\sum r_i$ and $\sum \gamma_i$.
- (d) Conditioning on K ,

$$\mathbb{E} e^{tV} = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} (1 - 2t)^{-(r+2k)/2} = (1 - 2t)^{-r/2} e^{-\gamma/2} \exp\left(\frac{\gamma/2}{1 - 2t}\right),$$

and $-\frac{\gamma}{2} + \frac{\gamma}{2(1-2t)} = \frac{\gamma t}{1-2t}$. This is [Eq. 4.1.2](#). The density of a mixture is the mixture of the densities.

■

The mixture in (d) is more than a formula. It says that noncentrality acts like a random number of extra degrees of freedom, two at a time, with $\gamma/2$ of them on average. Since each extra pair of degrees of freedom pushes the distribution to the right, a larger γ should give a stochastically larger variable. That is the next result, and it is what makes tests based on these distributions sensible.

Example. (A noncentral chi-squared with four degrees of freedom)

Take $r = 4$ and $\gamma = 6$. The Poisson(3) weights of the central densities g_4, g_6, g_8, g_{10} in [Eq. 4.1.3](#) are 0.050, 0.149, 0.224 and 0.224, and the mean and variance are 10 and 32. The listing checks the mixture against a library implementation of the density. The script also squares normal vectors with a mean of length $\sqrt{6}$ in an arbitrary direction, and their average squared length is 10.002. A test that rejects when a $\chi^2(4)$ statistic exceeds its upper 5% point 9.488 rejects with probability 0.470 when the statistic is in fact $\chi^2(4, 6)$. [Figure 4.1.1](#) shows how the density moves right and spreads out as γ grows.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(20240401)

def ncx2_pdf_mixture(x, r, gamma, terms=200):
    """Density of chi^2(r, gamma) as a Poisson(gamma/2) mixture of central densities
    . """
    k = np.arange(terms)
    weights = stats.poisson.pmf(k, gamma / 2) # P(K = k)
    dens = stats.chi2.pdf(np.asarray(x)[: , None], r + 2 * k)
    return dens @ weights

r, gamma = 4, 6.0
x = np.linspace(0.05, 40, 400)
print(np.max(np.abs(ncx2_pdf_mixture(x, r, gamma) - stats.ncx2.pdf(x, r, gamma))))
```

[Run this code online](#)

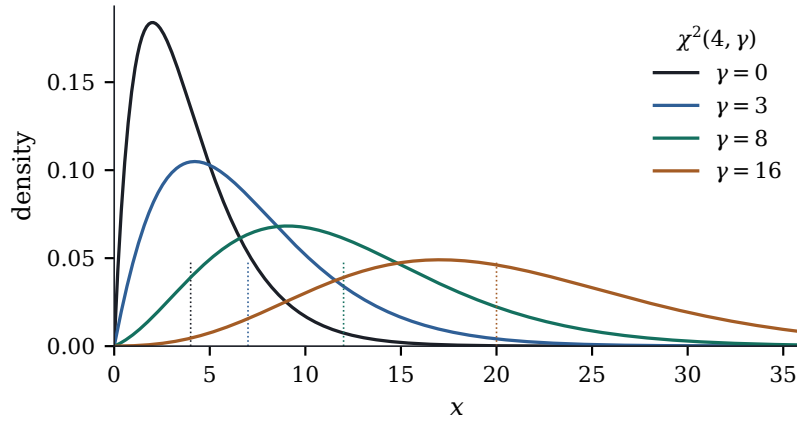


Figure 4.1.1: Densities of $\chi^2(4, \gamma)$. Dotted ticks mark the means $4 + \gamma$. The variance $2r + 4\gamma$ grows with the noncentrality, so the distributions flatten as they move right.

Proposition 4.1.3 — Monotonicity in the noncentrality

For fixed $r \geq 1$ and $x > 0$, the tail probability $\gamma \mapsto \Pr\{\chi^2(r, \gamma) > x\}$ is continuous and strictly increasing on $[0, \infty)$, and tends to 1 as $\gamma \rightarrow \infty$.

Proof. Let $h(k) = \Pr\{\chi^2(r + 2k) > x\}$ and $p_k(\lambda) = e^{-\lambda} \lambda^k / k!$. By Theorem 4.1.2(d), the tail probability is $\phi(\lambda) = \sum_k p_k(\lambda) h(k)$ with $\lambda = \gamma/2$. First, h is strictly increasing: if $V \sim \chi^2(r + 2k)$ and $E \sim \chi^2(2)$ are independent, then $V + E \sim \chi^2(r + 2k + 2)$ and $h(k + 1) - h(k) = \Pr(V \leq x < V + E) > 0$, since (V, E) has a positive joint density on $(0, \infty)^2$. Next, $p'_k(\lambda) = p_{k-1}(\lambda) - p_k(\lambda)$ (with $p_{-1} = 0$). Because $0 \leq h \leq 1$, the series of derivatives converges uniformly on bounded intervals, so term-by-term differentiation is valid and

$$\phi'(\lambda) = \sum_{k \geq 0} (p_{k-1}(\lambda) - p_k(\lambda)) h(k) = \sum_{k \geq 0} p_k(\lambda) (h(k + 1) - h(k)) > 0.$$

For the limit, fix $\varepsilon > 0$. Since $\chi^2(r + 2m)$ has mean $r + 2m$ and standard deviation $\sqrt{2r + 4m}$, Chebyshev's inequality gives $h(m) > 1 - \varepsilon$ for some m . Then $\phi(\lambda) \geq h(m) \Pr(K \geq m)$ for $K \sim \text{Poisson}(\lambda)$, and $\Pr(K \geq m) \rightarrow 1$ as $\lambda \rightarrow \infty$. ■

4.1.2 Noncentral F and t

Test statistics divide one quadratic form by another, to remove the unknown scale. The two ratio distributions that result are defined next.

Definition 4.1.2 — Noncentral F

Let $U \sim \chi^2(r, \gamma)$ and $V \sim \chi^2(s)$ be independent, $r, s \geq 1$. The law of

$$F = \frac{U/r}{V/s}$$

is the **noncentral F distribution** $F(r, s, \gamma)$, with numerator and denominator degrees of freedom r and s and noncentrality γ . The central $F(r, s) = F(r, s, 0)$ has upper α point $F_\alpha(r, s)$.

Definition 4.1.3 — Noncentral t

Let $X \sim N(\delta, 1)$ and $V \sim \chi^2(s)$ be independent. The law of $T = X/\sqrt{V/s}$ is the **noncentral t distribution** $t(s, \delta)$, and $t(s) = t(s, 0)$.

Only the numerator is allowed to be noncentral. A noncentral denominator gives a *doubly noncentral* F , which appears when the model used for the error sum of squares is itself wrong (Exercise (B3)). Two relations are immediate. First, $X^2 \sim \chi^2(1, \delta^2)$, so $T^2 \sim F(1, s, \delta^2)$: a two-sided t test is an F test with one numerator degree of freedom. Second, since $E(V^{-1}) = 1/(s-2)$ for $s > 2$, independence gives

$$E(F) = \frac{r + \gamma}{r} \cdot \frac{s}{s - 2}, \quad s > 2.$$

The central F is the null distribution of the tests of Chapter 11, and $F(r, s, \gamma)$ is their distribution under an alternative. The power of such a test is therefore $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$, and the following fact is what guarantees that it behaves as a test should.

Theorem 4.1.4 — Power of the F test increases with γ

For fixed $r, s \geq 1$ and $c > 0$, $\beta(\gamma) = \Pr\{F(r, s, \gamma) > c\}$ is continuous and strictly increasing in $\gamma \geq 0$, and $\beta(\gamma) \rightarrow 1$ as $\gamma \rightarrow \infty$. In particular, the level- α test that rejects when $F > F_\alpha(r, s)$ has power greater than α at every $\gamma > 0$.

Proof. With U, V as in Definition 4.1.2, $F > c$ iff $U > crV/s$. Condition on V , which is independent of U :

$$\beta(\gamma) = E[\psi_\gamma(crV/s)], \quad \psi_\gamma(x) = \Pr\{\chi^2(r, \gamma) > x\}.$$

Since $V > 0$ with probability one, Proposition 4.1.3 gives, for $\gamma < \gamma'$, $\psi_\gamma(crV/s) < \psi_{\gamma'}(crV/s)$ almost surely, so $\beta(\gamma) < \beta(\gamma')$. Continuity and the limit follow from the same proposition by dominated convergence, since $0 \leq \psi_\gamma \leq 1$. ■

Two further monotonicity facts hold for fixed $\gamma > 0$ and fixed level. The power decreases as the numerator degrees of freedom r grow, and increases as the denominator degrees of freedom s grow (Ghosh 1973). The first says that spreading a fixed amount of signal over more dimensions makes it harder to detect. The second says that estimating σ^2 better helps. We use both only qualitatively, and Figure 4.4.1 in Section 4.4 shows the first. Because $T^2 \sim F(1, s, \delta^2)$, Theorem 4.1.4 also shows that the power of the two-sided t test increases with $|\delta|$ (Exercise

(A2)).

4.1.3 Exercises

4.1.3.1 A. Check your understanding

Exercise A1.

Show that $\chi^2(2)$ is the exponential distribution with mean 2. Use [Theorem 4.1.2\(d\)](#) to show that

$$\Pr\{\chi^2(2, \gamma) > x\} = \sum_{k \geq 0} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} e^{-x/2} \sum_{j=0}^k \frac{(x/2)^j}{j!}.$$

Exercise A2.

Show that $T \sim t(s, \delta)$ implies $T^2 \sim F(1, s, \delta^2)$. Deduce that the power of the two-sided level- α t test depends on δ only through $|\delta|$ and increases with it. Show directly, by conditioning on the denominator, that the power of the one-sided test that rejects for $T > c$ increases with δ .

Solution. $T^2 = X^2/(V/s)$ with $X^2 \sim \chi^2(1, \delta^2)$ independent of V , which is [Definition 4.1.2](#). The two-sided test rejects when $|T| > t_{\alpha/2}(s)$, that is, when $T^2 > F_{\alpha}(1, s)$. Its power is therefore $\Pr\{F(1, s, \delta^2) > F_{\alpha}(1, s)\}$, a strictly increasing function of δ^2 by [Theorem 4.1.4](#). For the one-sided test, $\Pr(T > c) = \mathbb{E} \Pr\{X > c\sqrt{V/s} \mid V\} = \mathbb{E} \Phi(\delta - c\sqrt{V/s})$, and Φ is strictly increasing.

4.1.3.2 B. Practice

Exercise B1.

Show that the k th cumulant of $\chi^2(r, \gamma)$ is $2^{k-1}(k-1)!(r+k\gamma)$. Deduce that its skewness tends to zero like $3/\sqrt{\gamma}$ as $\gamma \rightarrow \infty$ with r fixed.

Solution. Using $-\frac{1}{2} \log(1-2t) = \sum_{k \geq 1} (2t)^k / (2k)$ and $t/(1-2t) = \sum_{k \geq 1} 2^{k-1} t^k$, the cumulant generating function is

$$K(t) = \sum_{k \geq 1} t^k 2^{k-1} \left(\frac{r}{k} + \gamma \right) = \sum_{k \geq 1} \frac{t^k}{k!} 2^{k-1} (k-1)! (r+k\gamma).$$

The k th cumulant is $k!$ times the coefficient of t^k . In particular $\kappa_2 = 2(r+2\gamma)$ and $\kappa_3 = 8(r+3\gamma)$, so the skewness is $\kappa_3/\kappa_2^{3/2} = 8(r+3\gamma)/\{2(r+2\gamma)\}^{3/2}$. As $\gamma \rightarrow \infty$ this behaves like $24\gamma/(4\gamma)^{3/2} = 3/\sqrt{\gamma}$.

Exercise B2.

Show that $\chi^2(1, \gamma)$ has density $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} \cosh \sqrt{\gamma x}$ for $x > 0$. Verify that expanding $\cosh$ in a power series reproduces [Eq. 4.1.3](#). *Hint:* $\Gamma(k + \frac{1}{2}) = \sqrt{\pi} (2k)! / (4^k k!)$.

Solution. Let $X = (W + \sqrt{\gamma})^2$ with $W \sim N(0, 1)$. For $x > 0$, $X \leq x$ iff $-\sqrt{x} - \sqrt{\gamma} \leq W \leq \sqrt{x} - \sqrt{\gamma}$. Differentiating with ϕ the standard normal density,

$$f(x) = \frac{\phi(\sqrt{x} - \sqrt{\gamma}) + \phi(\sqrt{x} + \sqrt{\gamma})}{2\sqrt{x}} = \frac{e^{-(x+\gamma)/2}}{\sqrt{2\pi x}} \cdot \frac{e^{\sqrt{\gamma x}} + e^{-\sqrt{\gamma x}}}{2}.$$

Expanding, $f(x) = (2\pi x)^{-1/2} e^{-(x+\gamma)/2} \sum_k (\gamma x)^k / (2k)!$. The k th term of [Eq. 4.1.3](#) is

$$\frac{e^{-\gamma/2} (\gamma/2)^k x^{k-1/2} e^{-x/2}}{k! 2^{k+1/2} \Gamma(k + \frac{1}{2})}.$$

By the hint the denominator is $k! 2^{k+1/2} \sqrt{\pi} (2k)! / (4^k k!) = \sqrt{2\pi} 2^{-k} (2k)!$, and the k th term becomes $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} (\gamma x)^k / (2k)!$, as required.

Exercise B3.

Let $U_\gamma \sim \chi^2(r, \gamma)$ with r fixed. Show that $(U_\gamma - r - \gamma) / \sqrt{2r + 4\gamma}$ converges in distribution to $N(0, 1)$ as $\gamma \rightarrow \infty$. *Hint:* U_γ has the law of $(W_1 + \sqrt{\gamma})^2 + W_2^2 + \cdots + W_r^2$ with $W_1, \dots, W_r$ independent standard normal.

Exercise B4.

Let $U \sim \chi^2(a)$ and $V \sim \chi^2(b)$ be independent. Show that $B = U/(U + V)$ has the $\text{Beta}(a/2, b/2)$ distribution and is independent of $U + V$, and that $F = (U/a)/(V/b) = (b/a) B/(1 - B)$. Deduce the null distribution of R^2 asked for in [Chapter 6](#).

Solution. The joint density of (U, V) is proportional to $u^{a/2-1} v^{b/2-1} e^{-(u+v)/2}$. Change variables to $w = u/(u + v) \in (0, 1)$ and $s = u + v > 0$, so that $u = ws$, $v = (1 - w)s$ and the Jacobian is s . The density of (W, S) is proportional to

$$w^{a/2-1} (1 - w)^{b/2-1} \cdot s^{(a+b)/2-1} e^{-s/2},$$

a product of a function of w and a function of s . So B and $U + V$ are independent, $B \sim \text{Beta}(a/2, b/2)$ and $U + V \sim \chi^2(a + b)$. Also $U/V = B/(1 - B)$. For R^2 in a model with intercept, $R^2 = \text{SSR}/(\text{SSR} + \text{SSE})$, where $\text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{Y}\|^2$ and $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ come from orthogonal projections of ranks $r - 1$ and $n - r$. If $\boldsymbol{\mu} = \boldsymbol{\mu}\mathbf{1}$, both noncentralities vanish, and Theorem 4.4.4 with $a = r - 1$, $b = n - r$ gives $R^2 \sim \text{Beta}((r - 1)/2, (n - r)/2)$.

Exercise B5.

Show that $f_{r,\gamma}(x)/g_r(x)$ is strictly increasing in $x > 0$ for every $\gamma > 0$. Conclude that among tests based on one observation $U \sim \chi^2(r, \gamma)$ of $\gamma = 0$ against a fixed $\gamma > 0$, rejecting for large U is most powerful.